



Education

PhD | Biomedical Engineering

Rensselaer Polytechnic Institute
2012-2017

- Research: machine learning, image processing, brain imaging, surgical skill assessment
- Advisors: Suvranu De, Xavier Intes

MS | Biomedical Engineering

Rensselaer Polytechnic Institute
2010-2012

- Research: image processing, virtual simulators, surgical skill assessment

BS Biomedical Engineering

University of Minnesota - Twin Cities
2005-2009

Skills

Languages

Python • SQL • bash
Terraform • Matlab • C++ • R

Machine Learning

Open source frameworks

PyTorch • Tensorflow • hugging-face •
scikit-learn • pandas • numpy • scipy
catboost • airflow • XGBoost
dask • shap • sentence-transformers •
lifelines

Algorithms

deep learning • CNN • transformers
gradient boosting • contrastive-learning
clustering • regression • classification
survival • PCA / LDA • bayesian methods
model selection • model validation

Healthcare

Real world data (RWD) • EHR • ICD-9/10
Electrocardiograms • Echocardiograms
Medical imaging • Cardiology • Oncology

General

Databricks • AWS • GCP • Azure •
(yup, all three)
TeX • linux • UNIX • git • docker
spark • DBT • CUDA • OpenCV
Kubernetes • CI/CD



Experience

Senior Staff Machine Learning Engineer | Altana AI

Nov 2024 - Present

- Designed, validated, and deployed a narcotics screening classifier trained on 200M transactions (PR - AUC = **0.22 @ 0.00001** prevalence), running at **50ms p50 latency** and handling **10M inferences/week** for a Fortune 50 firm. **Patent pending.**
- Established org-wide MLOps best practices for cross-cloud integrations, databricks deployments, and CI/CD practices.

Senior Staff Machine Learning Scientist | Tempus AI

Aug 2019 - Oct 2024

- Co-invented the **FDA 510(k) approved** Tempus Atrial Fibrillation (AF) algorithm aimed to predict first time AF risk within one year. Deep learning model is trained on **3.5M ECGs from 2M patients** (ROC - AUC = **0.84**) deployed in **10+** hospitals.
- Inventor and maintainer of the Tempus ECG training platform, a distributed cloud-based GPU modeling framework that powers all multi-modal research and cardio ECG production models, generating **\$20M+** in revenue.
- Lead data engineer for Tempus Cardio with responsibilities ranging from creating databases for EHR and imaging data and successful integration of four healthcare partners consisting of **2.5M+ patients and 10TB+ worth of data** enabling data access to **10+** team members.
- Validated and implemented a de-identification pipeline to scalably de-identify **1 billion** clinical notes from **2.5M** patients within **40** hours yielding **98%** sensitivity.
- Led org-wide documentation protocols to establish best practices on coding standards, cloud computing fundamentals, and clinical validation methods.
- Managed four machine learning scientists to execute on cross-functional business initiatives.

Founder and Chief Scientist | Draycon Labs

Jul 2019 - Present

- Built, validated, and deployed an automated, gigapixel histology slide segmentation based on deep learning on Azure for end-end slide analysis workflows for Fortune 500 medical device firms with **100%** license renewal rate for **6 years**.

Senior Data Scientist | Food Genius (acquired by US Foods)

Jul 2018 - Jul 2019

- Designed, built, and deployed a full-stack, machine-learning based web app that predicts supply chain service levels enterprise wide, with **36%** higher balanced accuracy than food industry standards

Key Publications and Patents

Articles

- Prediction of mortality from 12-lead electrocardiogram voltage data using a deep neural network *Nature Medicine*
- Deep Neural Networks Can Predict New-Onset Atrial Fibrillation From the 12-Lead ECG and Help Identify Those at Risk of Atrial Fibrillation-Related Stroke *Circulation*
- Assessing bimanual motor skills with optical neuroimaging *Science Advances*

Patents

- US11869668B2, US20230245782A1, US11657921B2, US20230028783A1, US20210076960A1